

Generation Interconnection System Impact Study Report

For Queue Project Q17-0835

Red Hill 138 kV
January 18, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0835. The Interconnection Customer proposes a 95.5 MW solar photovoltaic facility interconnecting at the Red Hill 138 kV substation in the Virginia area. The request was received on July 23, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 315 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,216,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Amber – Bremono Bluff 138 KV	Lake Lure – Bluff City 138 KV	122.4	194	6.3
Colgate – Albreda Tans Mtn Tap 138 KV Line	Louisa Generation Facility – Fluvanna P.S 345 KV	108.4	228	6.8
Colgate – Proffit 138 KV	Louisa Generation Facility – Fluvanna P.S 345 KV	104.8	202	6.8
Gordonsville – Proffit Substation 138 KV Line	Fluvanna P.S – Louisa Generation Facility 345 KV L	102.2	191	6.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bremono Bluff – Amber 138 KV	Rebuild line with higher-capacity conductor	\$12,282,000	11% (\$1,363,000)
Colgate – Albreda Tans Mtn Tap 138 KV Line	Reconductor line	\$1,952,000	9% (\$175,000)
Proffit – Colgate 138 KV	Reconductor line	\$3,025,000	9% (\$272,000)
Gordonsville – Proffit Substation 138 KV	Reconductor line	\$4,522,000	9% (\$406,000)
	Total allocated to project		\$2,216,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.